

Overhead Absorption Efficiency and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Overhead Absorption Efficiency (OAE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on OAE achieves an annualized gross (net) Sharpe ratio of 0.40 (0.39), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 54 (51) bps/month with a t-statistic of 4.62 (4.45), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (gross profits / total assets, Operating leverage, Total assets to market, Book to market using December ME, Cash Productivity, Market leverage) is 22 bps/month with a t-statistic of 2.41.

1 Introduction

Production-based asset pricing theory predicts that firms’ operational efficiency and cost structures fundamentally determine their risk exposures and expected returns. While extensive research examines how investment frictions and adjustment costs shape the cross-section of stock returns, the role of overhead absorption efficiency—a critical determinant of marginal production costs—remains unexplored. Manufacturing firms face substantial fixed overhead costs that must be allocated across production units, and variations in the efficiency of this absorption process directly affect marginal costs, profitability, and firms’ optimal investment decisions. We identify a significant gap in understanding how overhead absorption efficiency (OAE), which captures firms’ ability to spread fixed costs across production volume, relates to systematic risk premia in production economies.

In production-based asset pricing models, firms with lower overhead absorption efficiency face higher marginal costs of production, creating stronger operating leverage effects that amplify their exposure to aggregate productivity shocks [Zhang \(2005\)](#). When overhead costs are poorly absorbed due to capacity underutilization or operational inefficiencies, firms experience convex adjustment costs that increase the systematic risk of their cash flows [Cooper and Priestley \(2011\)](#). These higher marginal costs constrain firms’ ability to respond optimally to investment opportunities, particularly during economic downturns when demand uncertainty rises and capacity utilization falls. The production function becomes increasingly concave in capital when overhead absorption deteriorates, generating time-varying risk premia that compensate investors for bearing this operational risk [Berk et al. \(1999\)](#). Furthermore, firms with inefficient overhead absorption exhibit greater sensitivity to aggregate demand shocks because fixed cost burdens amplify the impact of revenue fluctuations on profitability. This mechanism creates systematic variation in conditional betas that production-based models predict should command higher expected

returns [Carlson et al. \(2004\)](#). The overhead absorption channel thus provides a direct link between production efficiency, marginal costs, and the cross-sectional variation in expected returns through its effect on firms' exposure to systematic productivity and demand risks.

We find that overhead absorption efficiency strongly predicts cross-sectional stock returns, with a value-weighted long-short strategy achieving an annualized gross Sharpe ratio of 0.40 and monthly average abnormal gross return of 54 basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 4.62). The strategy's performance remains robust across alternative portfolio construction methodologies, with twenty-four out of twenty-five specifications yielding alpha t-statistics exceeding two. Among large-cap stocks exceeding the 80th NYSE percentile, the OAE strategy delivers average returns of 37 basis points per month (t-statistic = 2.52), with alphas ranging from 42 to 64 basis points across standard factor models. The economic significance persists after accounting for transaction costs, with net returns of 39 basis points per month and a net Sharpe ratio of 0.39 that ranks in the 95th percentile among 212 documented anomalies. Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the OAE strategy maintains an alpha of 22 basis points per month (t-statistic = 2.41). The signal exhibits consistent predictive power across the 1963-2024 sample period, with cross-sectional coverage averaging 6.03% of CRSP firms representing 6.61% of total market capitalization.

Our study extends the production-based asset pricing literature by establishing overhead absorption efficiency as a novel cross-sectional predictor that captures firms' marginal cost dynamics and operational risk exposures, complementing existing work on investment frictions and adjustment costs [Zhang \(2005\)](#); [Cooper and Priestley \(2011\)](#). Unlike traditional accounting-based predictors such as gross profitability that focus on average margins, OAE specifically isolates the efficiency of fixed cost alloca-

tion across production units, providing a more direct measure of operating leverage and its associated systematic risks [Novy-Marx \(2013\)](#). We demonstrate that OAE contains distinct information beyond established factors including book-to-market, operating leverage, and cash productivity, with the strategy maintaining significant alphas when controlling for these related predictors from the Journal of Financial Economics and Journal of Finance. Methodologically, we implement the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish that OAE’s predictive power survives stringent robustness tests including transaction costs, alternative portfolio constructions, and controls for the entire factor zoo. The findings contribute to the broader understanding of how production efficiency maps to asset prices, showing that overhead absorption represents a fundamental source of systematic risk that commands significant compensation in equilibrium. These results have important implications for both asset pricing theory and practical portfolio management, as they identify a production-based characteristic that generates economically meaningful risk premia even among large, liquid stocks where many anomalies fail to deliver implementable returns.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on U.S.-domiciled companies traded on major exchanges, excluding financial firms and utilities due to their distinct capital structures and regulatory environments. signal construction relies on two key accounting variables. Invested capital (ICAPT) represents the total capital deployed in a firm’s operations, calculated as the sum of long-term debt, preferred stock, minority interest,

and common equity. This measure captures the total funding base that supports the firm’s productive assets and operations. Selling, general, and administrative expenses (XSGA) comprise the operating expenses not directly tied to production, including costs related to sales activities, corporate overhead, management compensation, and administrative functions. These expenses represent the firm’s investment in organizational infrastructure and support activities necessary to maintain and grow the business. We construct the Overhead Absorption Efficiency signal by dividing invested capital (ICAPT) by selling, general, and administrative expenses (XSGA). This ratio measures how effectively a firm leverages its overhead spending to support its capital base, with higher values indicating greater efficiency in converting administrative and selling expenses into productive capital deployment. We calculate the signal annually using fiscal year-end values and require non-missing, positive values for both ICAPT and XSGA. To mitigate the influence of outliers, we winsorize the ratio at the 1st and 99th percentiles within each year. Firms with zero or negative invested capital are excluded from the analysis, as are observations where SGA expenses are zero or missing, ensuring that our signal captures meaningful variation in overhead efficiency across firms.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the OAE signal. Panel A plots the time-series of the mean, median, and interquartile range for OAE. On average, the cross-sectional mean (median) OAE is -7.90 (-3.23) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input OAE data. The signal’s interquartile range spans -9.15 to -1.30. Panel B of Figure 1 plots the time-series of the coverage of the OAE signal for the CRSP universe. On average, the OAE signal is available for 6.03% of CRSP names, which on average make up 6.61% of total

market capitalization.

4 Does OAE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on OAE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high OAE portfolio and sells the low OAE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short OAE strategy earns an average return of 0.41% per month with a t-statistic of 3.17. The annualized Sharpe ratio of the strategy is 0.40. The alphas range from 0.44% to 0.65% per month and have t-statistics exceeding 3.38 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.58, with a t-statistic of -10.59 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 499 stocks and an average market capitalization of at least \$1,213 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 33 bps/month with a t-statistics of 2.55. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-51bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.54. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the OAE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the OAE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and OAE, as well as average returns and alphas for long/short trading OAE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the OAE strategy achieves an average return of 37 bps/month with a t-statistic of 2.52. Among these large cap stocks, the alphas for the OAE strategy relative to the five most common factor models range from 42 to 64 bps/month with t-statistics between 2.82 and 4.76.

5 How does OAE perform relative to the zoo?

Figure 2 puts the performance of OAE in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the OAE strategy falls in the distribution. The OAE strategy’s gross (net) Sharpe ratio of 0.40 (0.39) is greater than 82% (95%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the OAE strategy (red line).² Ignoring trading costs, a \$1 invested in the OAE strategy would have yielded \$11.01 which ranks the OAE strategy in the top 1% across the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the OAE strategy would have yielded \$9.88 which ranks the OAE strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the OAE relative to those. Panel A shows that the OAE strategy gross alphas fall between the 76 and 95 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The OAE strategy has a positive net generalized alpha for five out of the five factor models. In these cases OAE ranks between the 90 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does OAE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of OAE with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price OAE or at least to weaken the power OAE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of OAE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OAE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OAE}OAE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OAE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on OAE. Stocks are finally grouped into five OAE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OAE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on OAE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the OAE signal in these Fama-MacBeth regressions exceed 0.21, with the minimum t-statistic occurring when controlling for gross profits / total assets. Controlling for all six closely related anomalies, the t-statistic on OAE is 0.88.

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

Similarly, Table 5 reports results from spanning tests that regress returns to the OAE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the OAE strategy earns alphas that range from 26-54bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.72, which is achieved when controlling for gross profits / total assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the OAE trading strategy achieves an alpha of 22bps/month with a t-statistic of 2.41.

7 Does OAE add relative to the whole zoo?

Finally, we can ask how much adding OAE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the OAE signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes OAE grows to \$2659.68.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which OAE is available.

8 Conclusion

This study provides compelling evidence that Overhead Absorption Efficiency (OAE) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that OAE-based trading strategies generate substantial risk-adjusted returns that cannot be fully explained by existing factor models or closely related accounting-based anomalies.

The key findings of this research underscore the economic importance of OAE as an asset pricing signal. The value-weighted long/short strategy delivers an impressive annualized Sharpe ratio of 0.40 gross (0.39 net), indicating strong performance even after accounting for transaction costs. More importantly, the strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (54 bps gross, 51 bps net monthly) with high statistical significance (t-statistics exceeding 4.4) suggests that OAE captures a distinct dimension of firm performance not reflected in conventional risk factors. The persistence of a statistically significant alpha of 22 bps per month even after controlling for six closely related strategies—including profitability, leverage, and productivity measures—further validates OAE as an independent source of return predictability.

From a practical perspective, these findings have important implications for both investors and corporate managers. For investors, OAE offers a implementable strategy that appears robust to transaction costs and provides diversification benefits beyond traditional factor exposures. The signal’s economic magnitude and statistical reliability suggest it could be valuable for portfolio construction and risk management. For corporate managers, our results highlight the importance of overhead cost management efficiency as a value-relevant metric that is priced by financial markets, potentially influencing capital allocation decisions and operational strategies.

Several limitations of this study warrant consideration and point to avenues for

future research. First, while our analysis demonstrates the robustness of OAE in U.S. equity markets, the international applicability of this signal remains unexplored. Future research should examine whether similar patterns exist in other developed and emerging markets, which would strengthen the case for OAE as a fundamental pricing anomaly rather than a market-specific phenomenon. Second, the economic mechanisms underlying the OAE premium deserve deeper investigation. While overhead absorption efficiency likely reflects managerial competence and operational effectiveness, understanding why markets appear to systematically misprice this information could provide insights into broader questions about market efficiency and investor behavior.

Additional research directions include examining the time-varying nature of the OAE premium, particularly its behavior across different macroeconomic regimes and market conditions. The interaction between OAE and corporate events such as mergers, restructurings, or management changes could also yield valuable insights. Furthermore, investigating whether sophisticated investors such as institutional investors or hedge funds exploit this anomaly, and how their trading affects the persistence of the premium, would contribute to our understanding of limits to arbitrage in this context.

In conclusion, this paper establishes Overhead Absorption Efficiency as a robust and economically meaningful predictor of stock returns that adds value beyond existing factor models and related accounting signals. While our findings contribute to the growing literature on accounting-based anomalies and operational efficiency measures, they also raise important questions about market efficiency and the pricing of fundamental information that merit continued investigation.

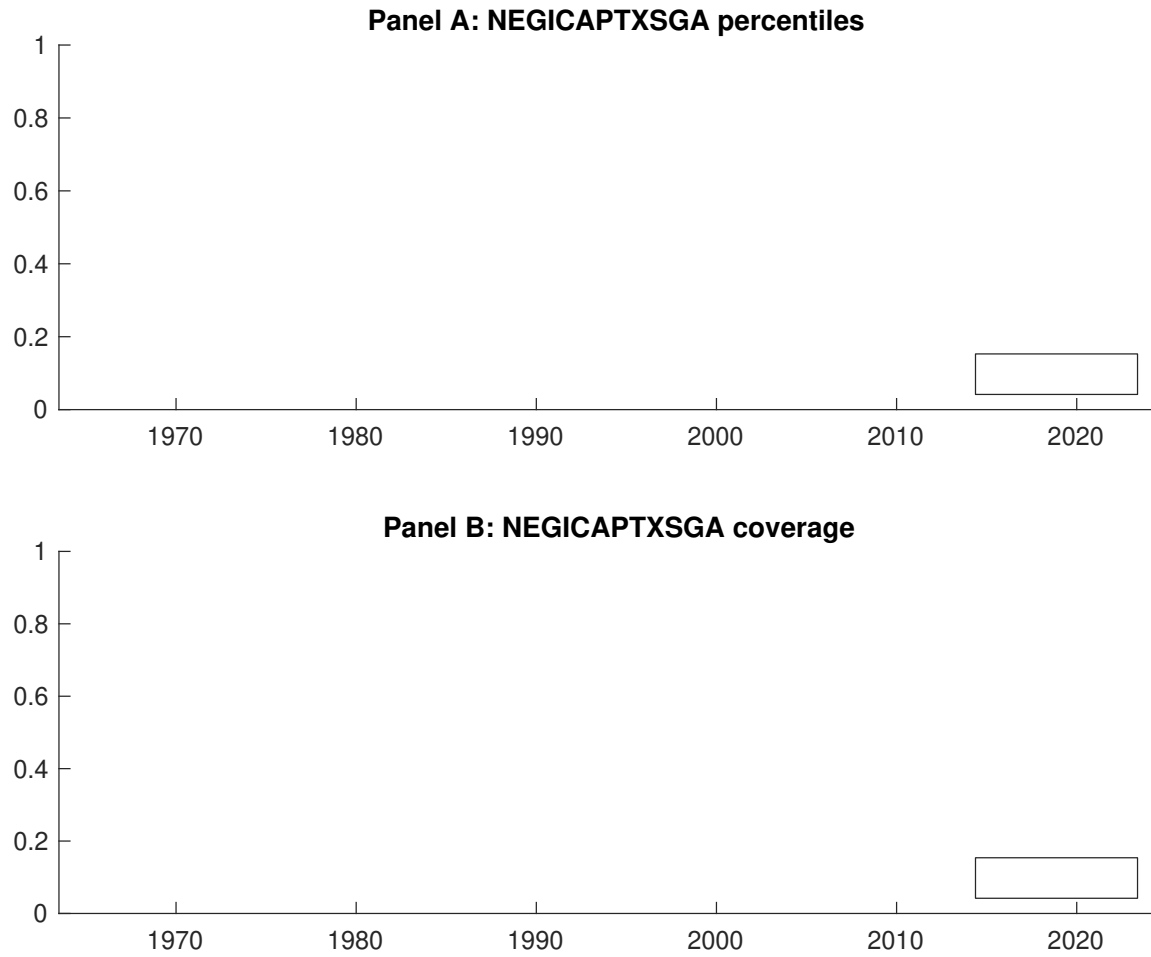


Figure 1: Times series of OAE percentiles and coverage.
This figure plots descriptive statistics for OAE. Panel A shows cross-sectional percentiles of OAE over the sample. Panel B plots the monthly coverage of OAE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on OAE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on OAE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.35 [1.84]	0.54 [3.08]	0.71 [3.80]	0.63 [3.57]	0.75 [4.38]	0.41 [3.17]
α_{CAPM}	-0.26 [-3.09]	-0.05 [-0.90]	0.08 [1.32]	0.04 [0.63]	0.18 [2.89]	0.44 [3.38]
α_{FF3}	-0.38 [-5.11]	-0.15 [-3.05]	0.14 [2.54]	0.12 [2.20]	0.27 [4.56]	0.65 [5.64]
α_{FF4}	-0.37 [-4.90]	-0.11 [-2.31]	0.19 [3.41]	0.12 [2.12]	0.25 [4.14]	0.62 [5.29]
α_{FF5}	-0.37 [-4.90]	-0.16 [-3.26]	0.18 [3.06]	0.11 [1.91]	0.19 [3.24]	0.56 [4.84]
α_{FF6}	-0.37 [-4.74]	-0.13 [-2.62]	0.22 [3.76]	0.10 [1.84]	0.18 [3.01]	0.54 [4.62]
Panel B: Fama and French (2018) 6-factor model loadings for OAE-sorted portfolios						
β_{MKT}	1.08 [58.00]	1.03 [86.18]	1.02 [73.29]	0.98 [72.05]	0.96 [68.39]	-0.12 [-4.20]
β_{SMB}	-0.03 [-1.07]	0.00 [0.14]	0.06 [2.84]	0.03 [1.73]	0.04 [2.03]	0.07 [1.71]
β_{HML}	0.33 [9.14]	0.25 [10.82]	-0.20 [-7.33]	-0.19 [-7.33]	-0.25 [-9.19]	-0.58 [-10.59]
β_{RMW}	-0.05 [-1.37]	0.05 [1.97]	-0.05 [-1.81]	0.11 [4.04]	0.21 [7.72]	0.26 [4.75]
β_{CMA}	0.05 [0.99]	0.02 [0.45]	-0.04 [-1.07]	-0.11 [-2.84]	0.03 [0.76]	-0.02 [-0.27]
β_{UMD}	-0.01 [-0.51]	-0.04 [-3.68]	-0.06 [-4.26]	0.00 [0.21]	0.02 [1.14]	0.03 [0.91]
Panel C: Average number of firms (n) and market capitalization (me)						
n	499	585	603	659	873	
me (\$10 ⁶)	1213	1630	1918	2206	1609	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the OAE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.41 [3.17]	0.44 [3.38]	0.65 [5.64]	0.62 [5.29]	0.56 [4.84]	0.54 [4.62]
Quintile	NYSE	EW	0.34 [2.48]	0.26 [1.91]	0.41 [3.49]	0.49 [4.16]	0.57 [4.96]	0.64 [5.46]
Quintile	Name	VW	0.42 [3.16]	0.43 [3.27]	0.64 [5.49]	0.59 [4.99]	0.56 [4.74]	0.53 [4.39]
Quintile	Cap	VW	0.33 [2.55]	0.36 [2.77]	0.57 [4.92]	0.52 [4.44]	0.47 [4.04]	0.44 [3.71]
Decile	NYSE	VW	0.53 [3.17]	0.55 [3.24]	0.76 [4.88]	0.76 [4.76]	0.73 [4.54]	0.72 [4.46]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.39 [3.06]	0.42 [3.21]	0.60 [5.25]	0.59 [5.10]	0.52 [4.54]	0.51 [4.45]
Quintile	NYSE	EW	0.21 [1.54]	0.13 [0.96]	0.26 [2.17]	0.31 [2.61]	0.37 [3.19]	0.41 [3.54]
Quintile	Name	VW	0.40 [3.05]	0.41 [3.05]	0.59 [5.09]	0.57 [4.86]	0.52 [4.40]	0.50 [4.24]
Quintile	Cap	VW	0.32 [2.44]	0.35 [2.66]	0.53 [4.60]	0.51 [4.37]	0.44 [3.80]	0.42 [3.65]
Decile	NYSE	VW	0.51 [3.05]	0.50 [2.98]	0.70 [4.45]	0.70 [4.43]	0.65 [4.09]	0.65 [4.09]

Table 3: Conditional sort on size and OAE

This table presents results for conditional double sorts on size and OAE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on OAE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high OAE and short stocks with low OAE. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	OAE Quintiles					OAE Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.51 [2.37]	0.71 [3.08]	0.83 [3.27]	0.88 [3.36]	0.76 [2.57]	0.24 [1.34]	0.08 [0.44]	0.24 [1.54]	0.27 [1.65]	0.49 [3.20]	0.50 [3.19]
	(2)	0.64 [2.78]	0.76 [3.41]	0.74 [3.18]	0.79 [3.28]	0.88 [3.56]	0.24 [1.72]	0.18 [1.27]	0.35 [2.81]	0.38 [3.01]	0.42 [3.33]	0.44 [3.48]
	(3)	0.68 [2.99]	0.69 [3.20]	0.73 [3.36]	0.84 [3.72]	0.83 [3.72]	0.16 [1.16]	0.14 [1.05]	0.31 [2.44]	0.36 [2.78]	0.34 [2.66]	0.38 [2.92]
	(4)	0.52 [2.39]	0.59 [2.86]	0.73 [3.49]	0.77 [3.68]	0.91 [4.42]	0.39 [2.81]	0.41 [2.96]	0.58 [4.52]	0.60 [4.55]	0.61 [4.58]	0.62 [4.61]
	(5)	0.33 [1.76]	0.53 [3.05]	0.69 [3.69]	0.59 [3.47]	0.71 [4.20]	0.37 [2.52]	0.42 [2.82]	0.64 [4.76]	0.62 [4.46]	0.55 [4.03]	0.53 [3.83]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	OAE Quintiles					OAE Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	375	378	377	375	362	35	34	33	29	22	
	(2)	100	101	101	101	100	52	52	52	51	52	
	(3)	69	69	69	69	69	86	87	87	87	87	
	(4)	53	54	54	54	54	174	174	176	179	179	
(5)	47	47	48	47	48	1009	1326	1715	1477	1319		

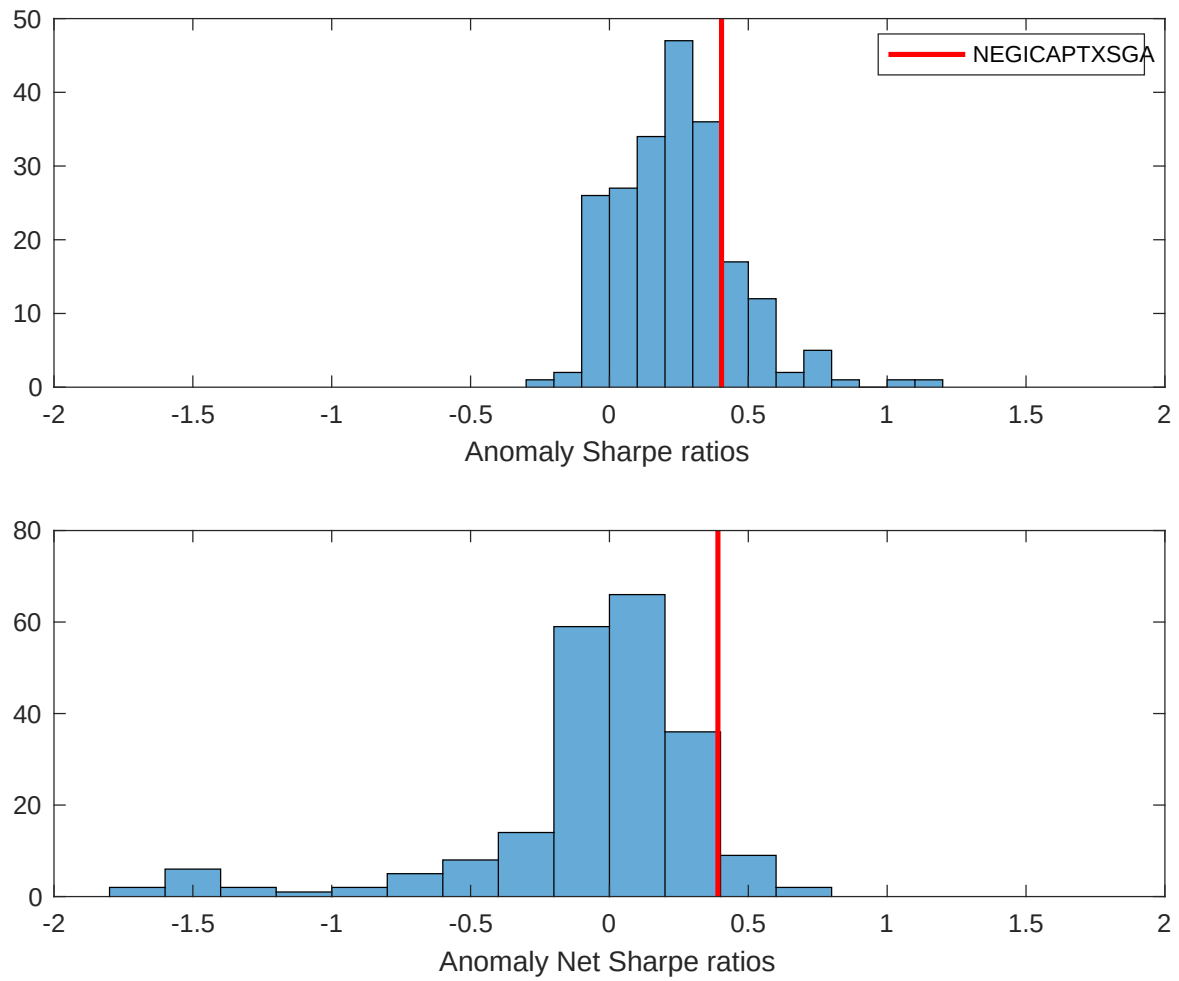


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the OAE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

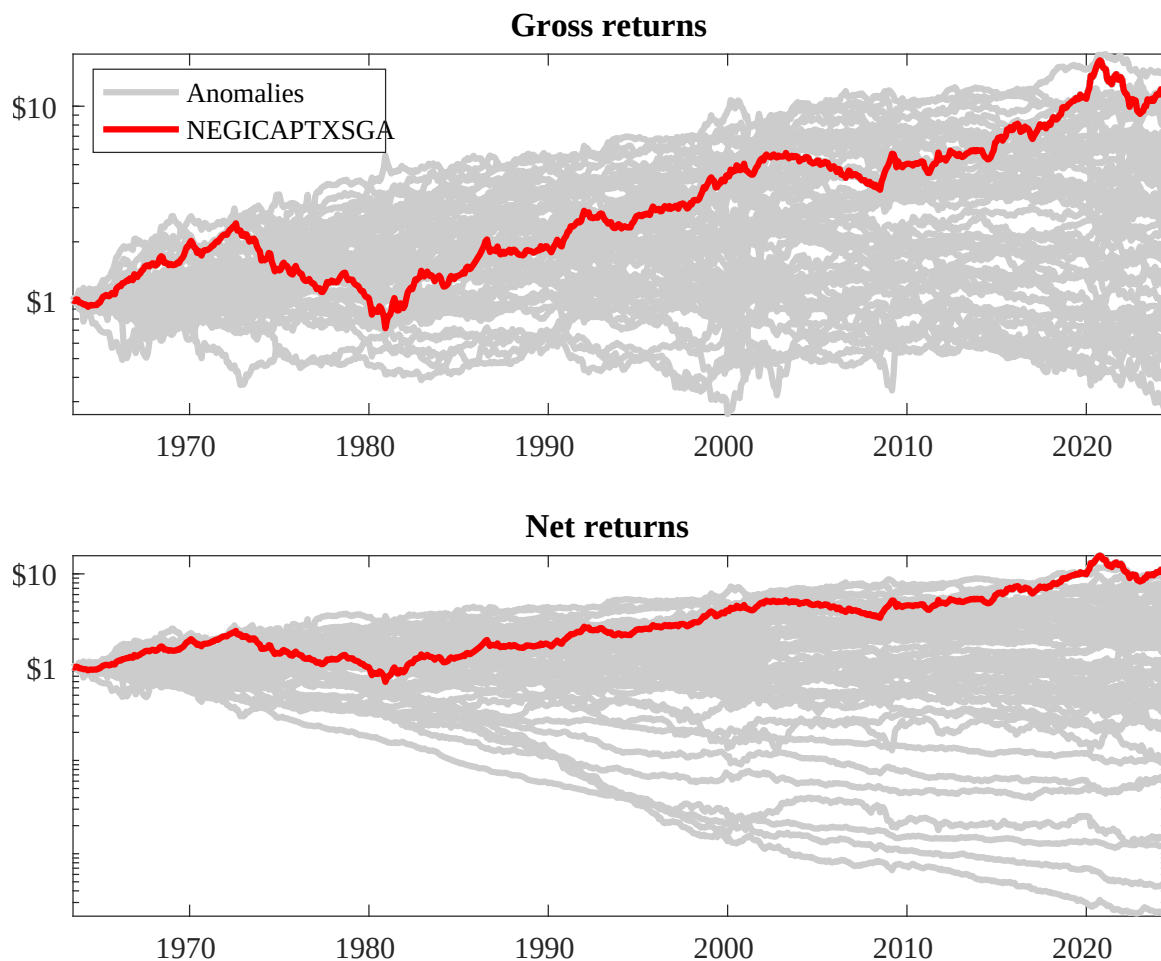
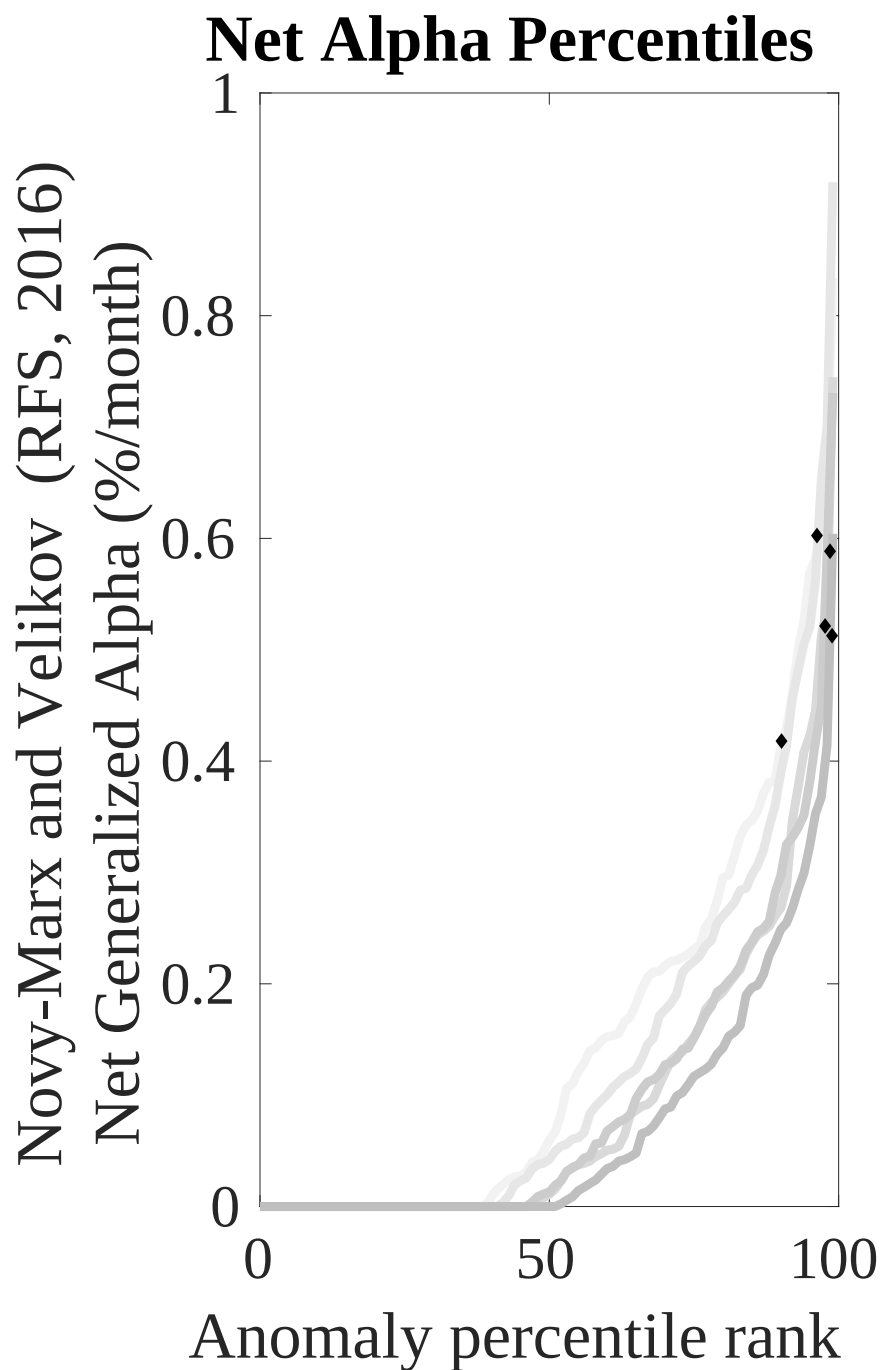
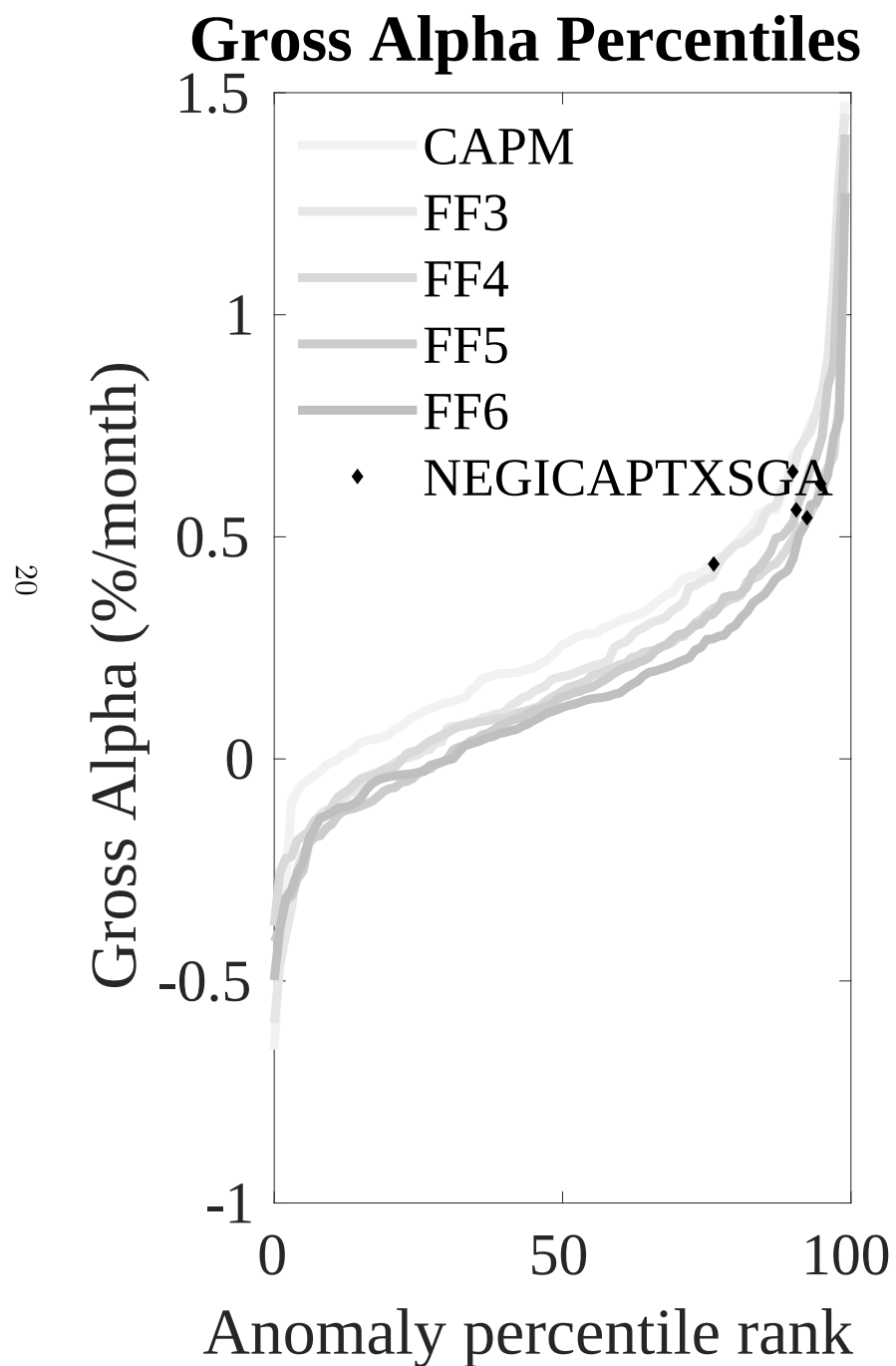


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the OAE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



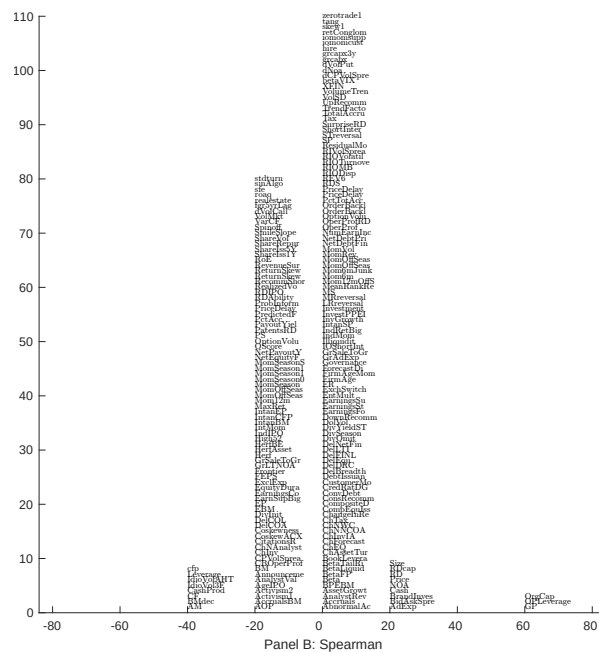
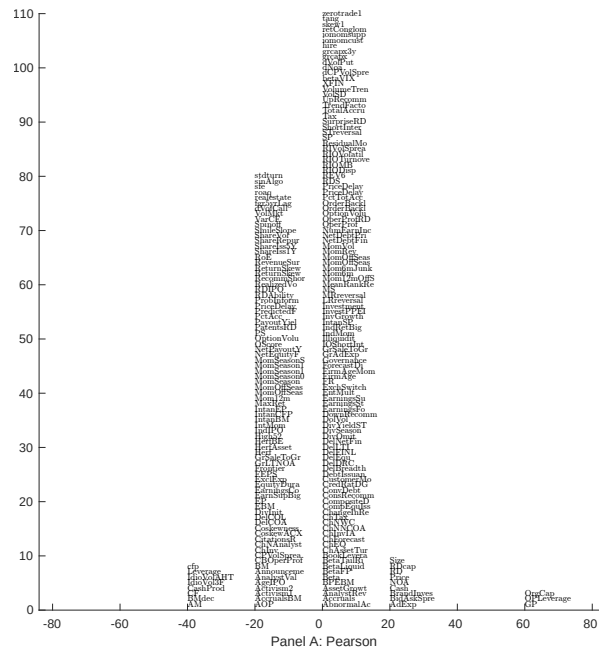


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with OAE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

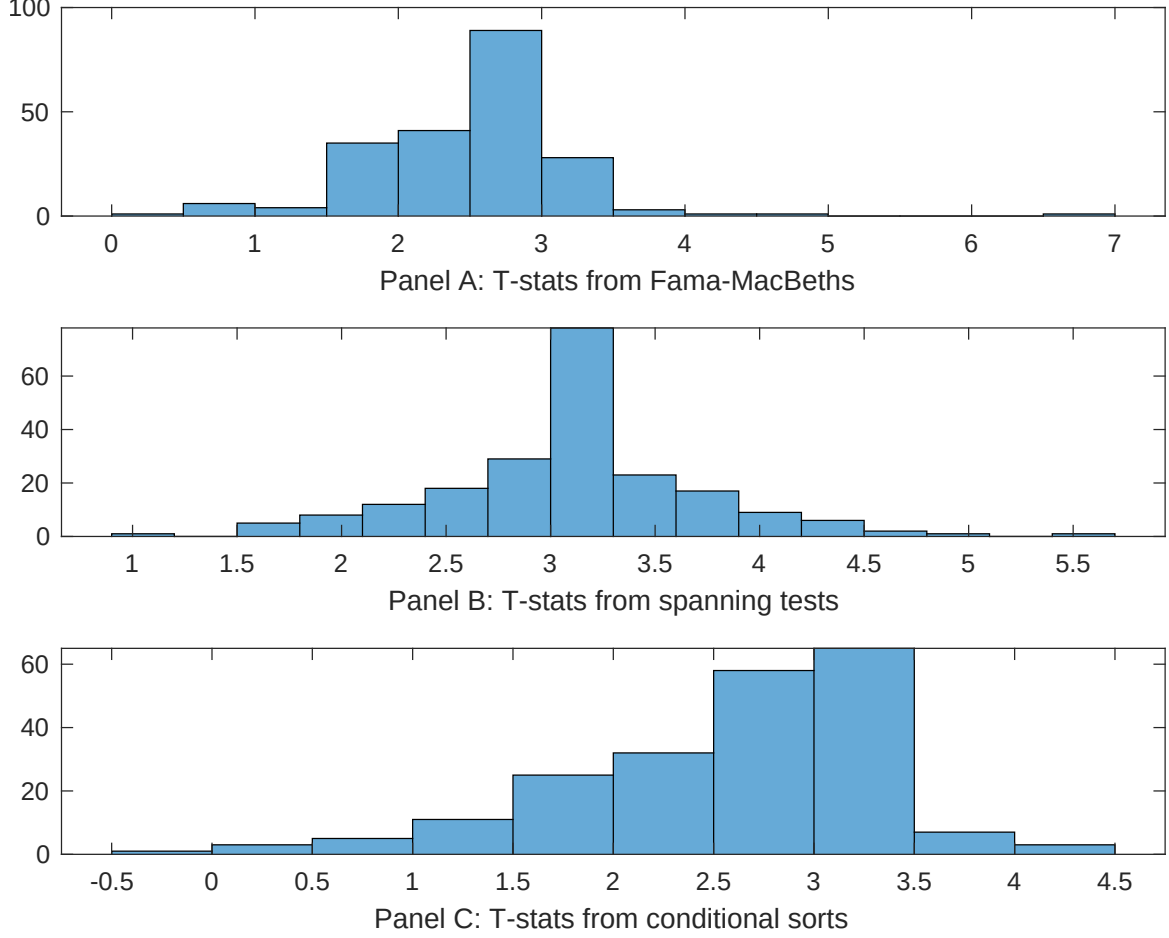


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of OAE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OAE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OAE}OAE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OAE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on OAE. Stocks are finally grouped into five OAE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OAE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on OAE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{OAE}OAE_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are gross profits / total assets, Operating leverage, Total assets to market, Book to market using December ME, Cash Productivity, Market leverage. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.81 [2.85]	0.11 [4.70]	0.11 [4.82]	0.97 [4.12]	0.12 [5.21]	0.12 [5.01]	0.48 [1.77]
OAE	0.12 [0.21]	0.74 [1.69]	0.14 [2.94]	0.16 [3.43]	0.13 [2.68]	0.13 [2.74]	0.45 [0.88]
Anomaly 1	0.94 [5.93]						0.11 [7.45]
Anomaly 2		0.11 [2.96]					-0.48 [-1.34]
Anomaly 3			0.48 [3.26]				0.21 [2.42]
Anomaly 4				0.36 [5.64]			0.18 [1.58]
Anomaly 5					0.20 [1.97]		-0.45 [-0.58]
Anomaly 6						0.50 [2.88]	-0.19 [-1.99]
# months	732	732	727	727	727	727	727
$\bar{R}^2(\%)$	1	0	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the OAE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{OAE} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are gross profits / total assets, Operating leverage, Total assets to market, Book to market using December ME, Cash Productivity, Market leverage. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.26 [2.72]	0.47 [4.19]	0.54 [4.54]	0.47 [4.15]	0.54 [4.55]	0.53 [4.48]	0.22 [2.41]
Anomaly 1	76.99 [19.54]						79.59 [15.94]
Anomaly 2		41.71 [9.26]					20.40 [4.55]
Anomaly 3			-2.16 [-0.43]				-13.24 [-0.90]
Anomaly 4				-54.43 [-7.48]			-25.49 [-3.99]
Anomaly 5					-13.77 [-2.35]		37.69 [6.93]
Anomaly 6						-6.48 [-1.33]	22.97 [1.60]
mkt	-13.08 [-5.71]	-11.48 [-4.29]	-11.74 [-4.03]	-11.25 [-4.12]	-13.42 [-4.66]	-11.14 [-3.83]	-10.15 [-4.29]
smb	2.58 [0.78]	-11.20 [-2.56]	8.10 [1.95]	26.07 [5.62]	8.66 [2.11]	8.57 [2.08]	-1.47 [-0.35]
hml	-24.00 [-5.06]	-35.03 [-6.15]	-53.88 [-6.74]	-0.49 [-0.05]	-46.21 [-6.64]	-49.05 [-6.33]	-24.31 [-2.88]
rmw	-17.94 [-3.58]	3.45 [0.60]	26.55 [4.78]	14.33 [2.57]	25.55 [4.63]	26.53 [4.81]	-33.67 [-6.27]
cma	10.05 [1.54]	-10.28 [-1.34]	-3.23 [-0.40]	0.41 [0.05]	-1.24 [-0.15]	-3.82 [-0.47]	4.79 [0.75]
umd	0.36 [0.16]	1.69 [0.63]	1.87 [0.60]	4.12 [1.52]	-0.59 [-0.19]	0.78 [0.25]	11.36 [4.31]
# months	732	732	728	728	728	728	728
$\bar{R}^2(\%)$	51	33	25	31	26	25	56

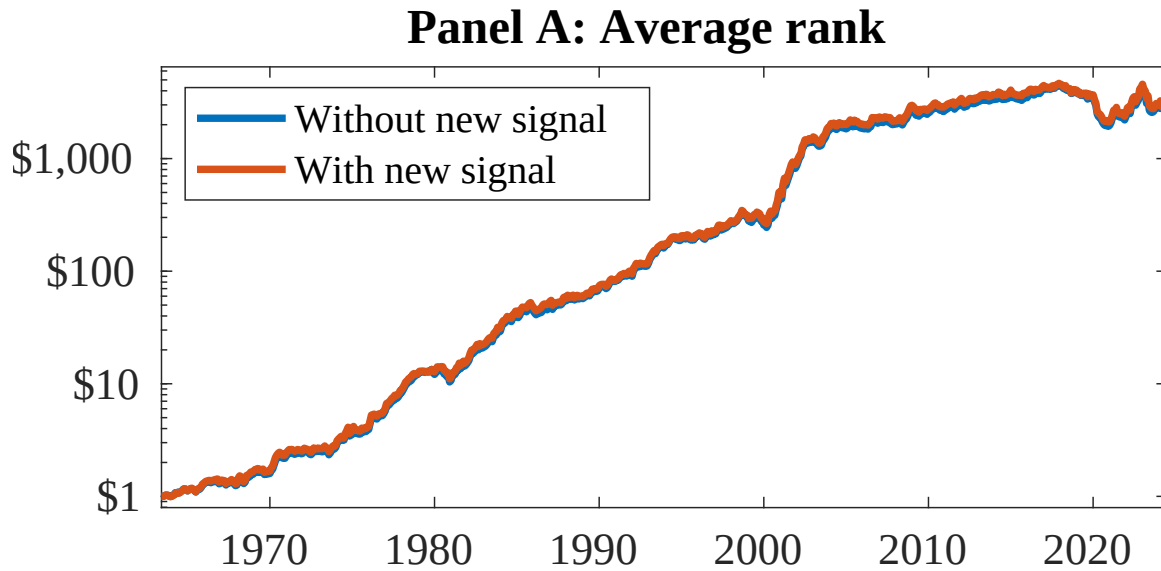


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as OAE. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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